



MICROSOFT INNOVATION CLUB, VIT CHENNAI
Circular Resource Intelligence: From Waste to Verified
Recovery
Sustainability & Climate

1. BACKGROUND

Cities, campuses, businesses, restaurants, retailers, and households discard large quantities of edible food, reusable goods, municipal waste, batteries, and electronic materials. Many of these resources still have useful value, yet they reach landfills because generation, collection, diagnosis, redistribution, repair, and recycling operate as *disconnected processes*.

Existing systems commonly depend on fixed collection schedules, manual calls, informal messaging groups, basic listing platforms, or one-time disposal decisions, and rarely combine urgency, location, material condition, safety, storage capacity, recipient demand, vehicle capacity, and processing availability.

The Problem

- Waste decisions are made without context — material may be discarded even when it can be reused, donated, repaired, repurposed, or recycled
- Collection is reactive or schedule-driven — vehicles visit low-demand locations while urgent or overflowing locations go unattended
- Supply and demand are poorly coordinated — donors, recipients, recyclers, volunteers, and collection teams can't see compatible needs and capacities in time

- Special waste streams require safe decisions — food expiry, battery health, hazardous behavior, contamination, and storage conditions can't be treated as ordinary listings
- Recovery is rarely verified — a pickup request or recommendation doesn't prove the resource reached a recipient or safe recovery facility

Target beneficiaries: municipalities, campuses, food providers, households, NGOs, repair centers, recyclers, waste workers, and local communities, with transport partners, regulators, sustainability teams, and circular-economy stakeholders as secondary beneficiaries.

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Sustainability & Climate — Problem Statement